

Discrete Math

Exercises:

- What does the mathematical statement means?

$$\forall x \in \mathbb{N}, x \geq 0$$

Solution:

For all x belongs to the set of natural numbers ($\mathbb{N}$), x is greater than or equal to 0.

- If set S contains 3 elements, how many elements are in ~~the~~ its power set $\mathcal{P}(S)$?

Solution:

$$2^3 = 8$$

$\Rightarrow$ 8 elements



If we have 10 pigeons and 9 nests,
and we want to place all the pigeons
into the nests, what does the
"Pigeonhole Principle" guarantee us?

Solution:

The pigeonhole principle guarantees that
at least one nest will have 2 or more
pigeons.

- What is the correct logical negation of the
following statement? and how to read it
in English?

$$\forall x, \exists y : P(x, y)$$

Solution:

$$\neg(\forall x, \exists y, P(x, y)) \equiv \exists x, \forall y : \neg P(x, y)$$

In English:

There exist an x , for all y
not hold $P(x, y)$.

- In mathematical logic, is the following conditional statement true or false:

"If $0 = 1$, then elephants can fly"? and why?

Solution:

The conditional statement is True

reasoning:

In mathematical logic, when the hypothesis is False, the conditional statement is considered vacuously true



- For a relation to be classified as an "equivalence relation", what are the three properties it must possess?

Solution:

An equivalence relation must be:

1- Reflexive

2- Symmetric

3- Transitive

Day 18 ☺

Never gave up

